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TypeTopology: a blackboard for research in constructive univalent mathematics

Martín Hötzel Escardó [1¶], **Tom de Jong** [2], **Ayberk Tosun** [3], **Chuangjie Xu** [4], **and Todd Waugh Ambridge** [1]

1 School of Computer Science, University of Birmingham, United Kingdom **2** Institute for Computing and Information Sciences, Radboud University, The Netherlands **3** Zeroth Research, United Kingdom (affiliated with **1** for the work described here) **4** Independent researcher, Germany ¶ Corresponding author

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Software

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Summary

TypeTopology is a constructive development of univalent mathematics and homotopy type theory (The Univalent Foundations Program, 2013) written in Agda (The Agda Development Team, 2025), started in 2010 and public since 2018. It currently has 1017 Agda files with 306K lines of code and comments, organized in about sixty subject directories, and it is checked with Agda 2.8.0 and 2.9.0. Among the mathematics it develops are the intrinsic topology of types (Escardó, 2013, 2026; Escardó & Streicher, 2016), injective types and their algebraic structure (de Jong & Escardó, 2027; Escardó, 2021), domain theory, (de Jong & Escardó, 2021, 2025), ordinals and their arithmetic (de Jong et al., 2023, 2026), locale theory (Arrieta et al., 2025), the Dedekind reals, models of set theory, higher-order games (Escardó & Oliva, 2023), and effectful forcing (Escardó et al., 2025). Four doctoral theses were written on material developed here, and their formalizations are part of TypeTopology, namely Chuangjie Xu's continuous computational interpretation of type theories (Xu, 2015), Tom de Jong's domain theory in constructive and predicative univalent foundations (de Jong, 2023), Todd Waugh Ambridge's exact real search, with optimization and regression over exact real numbers (Waugh Ambridge, 2024), and Ayberk Tosun's constructive and predicative locale theory (Tosun, 2025). The name TypeTopology was chosen both because of the well-known connection of topology with constructive mathematics (Smyth, 1992) and because homotopy theory is a branch of topology. The version described here is archived at (Escardó et al., 2026).

Statement of need

We wrote TypeTopology for mathematicians, logicians and computer scientists working in homotopy type theory and univalent foundations, with three goals in mind.

The first is a place to do new mathematics rather than to record mathematics already done. We use the development as a blackboard in which results appear and are discussed by the community before they are submitted for publication. Currently 35 publications and preprints have resulted from TypeTopology. A library organized as an encyclopedia of settled mathematics cannot play this role, because an encyclopedia records mathematics after it is understood, whereas a blackboard is where this understanding is developed. The blackboard material nevertheless evolves into a body of mathematics that others build on, organized by subject.

The second is mathematical exposition. The code is accompanied by extensive prose that explains the mathematics, discusses the choices made, and records what is known and what is open, so that a module can be read as a mathematical text. This is something readers and users of TypeTopology most often tell us they value.

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The third is precision about assumptions. Postulates are not allowed in TypeTopology, with the exception of a small experimental fragment, briefly discussed under *Software design* below. Function extensionality, propositional extensionality, the existence of propositional truncations, univalence, propositional resizing, and the principles of classical mathematics such as excluded middle, choice, global choice, LPO and WLPO, are all stated as explicit hypotheses of the theorems, or of the modules, that need them. A reader can therefore tell from the statement of a result exactly which assumptions it depends on, and we can tell which of our theorems are constructive and which are classical. Resizing is of particular interest, since its avoidance is what keeps the development predicative (de Jong & Escardó, 2023).

State of the field

Five actively maintained libraries of univalent mathematics are comparable in scope.

The UniMath library (Voevodsky et al., since 2013) is written in Rocq (The Rocq Development Team, 2026), formerly Coq, and goes back to Voevodsky (Voevodsky, 2015). It is built with the -type-in-type option, which makes the system inconsistent, and it assumes propositional resizing. Their own documentation records a succession of attempts to introduce resizing rules in order to avoid that option. We avoid propositional resizing to remain predicative.

The HoTT library (Bauer et al., 2017) is also written in Rocq, goes back to 2011, and has homotopy theory as its main subject. It is the closest to TypeTopology in its treatment of assumptions, since function extensionality, univalence and propositional resizing are axioms that a definition can use only by taking them as explicit hypotheses, so that their use is tracked. Higher inductive types, including propositional truncation, are built into the library with private inductive types and axioms (postulates). The use of the latter in theorems can be tracked using Rocq's Print Assumptions command.

The agda-unimath library (Rijke et al., since 2021) is written in Agda, as TypeTopology is, and is organized as an encyclopedia by mathematical subject. Its design principles record that it postulates function extensionality, univalence, the replacement axiom, the truncation operations, the interval, the circle, pushouts, extensionality of globular types, and the flat modality with propositional computation rules. Mathematics that needs excluded middle or choice is, by their own account, not yet developed there. TypeTopology, on the other hand, does not allow postulates, and studies classical principles deliberately, including the Cantor-Schröder-Bernstein theorem for arbitrary homotopy types under excluded middle, Zorn's lemma, the well-ordering principle, and a collection of taboos that measure how much classical logic a statement needs.

The Cubical library (The Cubical development team, since 2018) is the standard library for Cubical Agda (Vezzosi et al., 2021), an implementation of cubical type theory. In this library, univalence, function extensionality and the existence of higher inductive types all hold by construction. In TypeTopology, these are hypotheses of the theorems that use them.

The 1Lab (The 1Lab Development Team, since 2021) is a cross-linked reference resource for mathematics done in homotopy type theory, with category theory as its largest part. It is written in Mikan (Liao et al., since 2026), a fork of Agda made by its developers to support cubical type theory alone. The above remarks about the Cubical library also apply.

Software design

We deliberately work in a minimal version of intensional Martin-Löf type theory, which we call Spartan MLTT, and in a correspondingly minimal subset of Agda, which means that, in particular, we do not use Cubical Agda. The flags in typetopology.agda-lib fix the type theory as far as Agda at present allows, and they include, among others, --without-K to make the development compatible with homotopy type theory and univalent foundations. The

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development is indexed by `source/index.lagda`, which hereditarily imports every module that Agda can check as safe, and `source/AllModulesIndex.lagda` adds a few experimental modules that go beyond Martin-Löf type theory and univalent foundations. We define everything starting from Spartan MLTT rather than build on the Agda standard library (Daggitt et al., 2025), which serves the different purpose of supporting programming and proving in Agda at large, so Agda alone is needed to check the development.

The axioms required by a construction are hypotheses of its definition, or parameters of the module that collects them. We adopt the univalent point of view even in modules that assume no axioms, taking seriously the distinction between types that are singletons, propositions, sets, 1-groupoids, etc., and a large part of the development needs no axioms. We prefer Σ types to records, because univalent foundations supplies constructions and theorems for Σ types that we constantly apply.

Research impact statement

Currently 35 publications and preprints have arisen from TypeTopology, in the *Journal of Symbolic Logic*, *Mathematical Structures in Computer Science*, the *Annals of Pure and Applied Logic*, the *Journal of Pure and Applied Algebra*, *Logical Methods in Computer Science*, the *Journal of Homotopy and Related Structures*, and the proceedings of LICS, CSL, FSCD, TLCA and MFPS, by the authors and TypeTopology contributors, and the four doctoral theses named in the summary were written on material developed here. TypeTopology also has an entry in zbMATH Open (2026), which records the publications that cite it. Material from TypeTopology is also being reproduced in other developments, for example the theory of compact and searchable types in the HoTT library. TypeTopology is also used for teaching, incorporating the material of our Midlands Graduate School lecture notes (Escardó, 2019).

Testing

Running `agda AllModulesIndex.lagda` from the directory `source` type-checks every module, and the `Makefile` at the root does this with the released or the development version of Agda. A GitHub action runs the same check on every push and pull request. All safe modules are hereditarily imported by `source/index.lagda` and carry the option `--safe`, which rules out postulates, so a successful check is a computer verification that every theorem in them follows from the assumptions its statement declares. The safe fragment of TypeTopology is additionally checked with Mikan (ignoring its cubical features), as a double check.

AI usage disclosure

TypeTopology's file `AI-Policy.md` states that *"TypeTopology consists of literate Agda files that contain Agda code as well as accompanying prose. We will not consider AI generated Agda code or prose."* Regarding this paper, we wrote it from prose already available in TypeTopology, using AI assistance to bring it to the markdown format required by this journal.

Acknowledgements

TypeTopology has around forty contributors, listed in the README at GitHub. The authors of this paper are listed by amount of contribution to the development.

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